SEAM: SPEECH EMOTION-ALIGNED CONTROLLABLE MUSIC GENERATION

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ABSTRACT

Text-to-music diffusion models can generate high-quality instrumental audio, yet affective control remains largely text-driven and provides only coarse control over continuous emotion. We study *speech-conditioned affective music generation*, where expressive speech provides affect while an emotion-free text prompt specifies musical content. We propose SEAM, a lightweight conditioner for frozen Stable Audio 3 that maps speech into factorized valence and arousal tokens and injects them through dedicated mood cross-attention. SEAM further uses baseline-relative steering in speech emotion representation space to enable continuous affect control without emotion words in the text prompt. Automatic and human evaluations show improved speech–music affect alignment over same-backbone text conditioning and effective perceptual control of affect intensity. Audio samples are available at `https://jyshin0926.github.io/SEAM/`.

***Index Terms*—** speech emotion recognition, music generation, music mood, mood control

1. INTRODUCTION

Text-to-music generation has advanced from autoregressive codec-token models such as MusicGen [1] to latent diffusion systems including Stable Audio [2, 3, 4] and Stable Audio 3 [5]. These models can generate high-quality instrumental audio from natural-language prompts and provide an increasingly capable foundation for controllable music synthesis. As generation quality improves, affective control becomes important for applications in which music accompanies expressive speech, such as film, games, narration, and accessibility. In these settings, the desired musical affect may already be conveyed by a speaker's prosody, vocal intensity, and speaking style, making speech a natural reference for controlling the emotional character of background music [6, 7].

Existing text-to-music systems, however, remain predominantly text-driven: affect is typically specified through lexical descriptors such as "happy" or "sad." This interface provides only coarse control over a continuous affective space. The same emotion word may correspond to different valence and arousal levels across utterances, while modifiers such as "sadder" or "more excited" do not provide a well-defined continuous control signal. More importantly, lexical conditioning cannot directly exploit the fine-grained affective cues already present in expressive speech.

Prior emotion-conditioned music generation has commonly relied on categorical emotion labels or manually specified affect coordinates. For example, symbolic music models have used valence–arousal quadrants [8], emotion-dependent latent variables [9], and emotion labels for harmonization [10]. More recent approaches enable continuous affect control through disentangled valence–arousal conditioning [11] and continuous VA control for audio music generation [12, 13]. Fine-grained affect manipulation has also been studied in emotional TTS through activation steering [14, 15]. Although effective, these approaches require an explicit affect specification and therefore do not directly use expressive speech as the source of control.

Cross-modal speech–music research offers a complementary perspective: speech and music can be connected through shared affective representations. Doh et al. [16], for instance, align speech and music in a shared valence–arousal space for retrieval, while MEMGM [17] combines multimodal emotion representations for symbolic music generation. These studies show that low-dimensional affect can bridge speech and music. However, they do not address controllable audio music generation conditioned on the affect of a particular spoken utterance, while preserving independent textual control over musical content. This motivates *speech-conditioned affective music generation*: given a spoken utterance and an emotion-free instrument prompt, generate music whose valence–arousal (VA) follows the speech while leaving musical-content specification to text.

To this end, we propose SEAM, a lightweight speech-affect conditioner for frozen text-to-music diffusion models. SEAM factorizes speech affect into valence and arousal representations and injects them as mood tokens through dedicated mood cross-attention, separating affective control from musical-content prompting. It further introduces baseline-relative steering in speech emotion representation space, allowing affect intensity to be continuously adjusted without adding emotion words or explicit VA values to the text prompt. Unlike prior label- or coordinate-conditioned approaches, SEAM derives its affective control signal directly from expressive speech; unlike speech–music retrieval methods, it uses this signal to control the generation of new music without requiring naturally paired speech–music recordings.

Our contributions are:

- A speech-affect conditioner that separately represents valence and arousal as mood tokens and injects them into a frozen text-to-music DiT through dedicated mood cross-attention, while preserving the original text-conditioning pathway.
- An inference-time residual steering mechanism in the SER embedding space that enables continuous control of music affect using a speech-derived direction with adjustable strength α.
- An evaluation of speech–music affect alignment and continuous intensity control using both automatic VA estimation and human listening tests against same-backbone text-conditioning baselines.

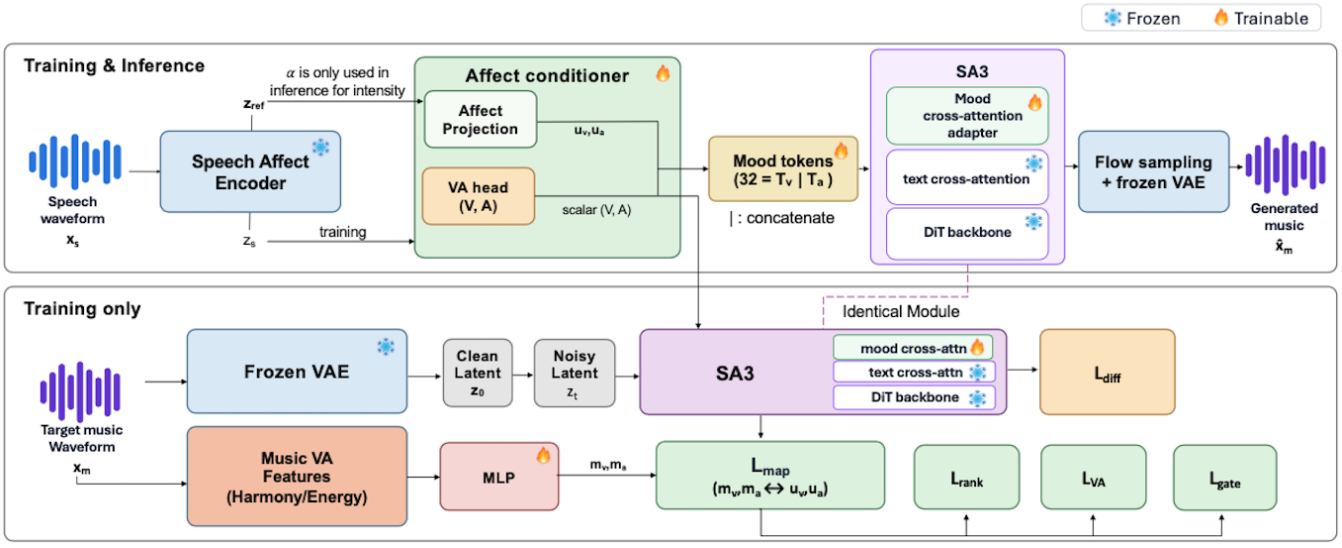


Fig. 1. Speech-conditioned affect injection into a frozen text-to-music DiT. The prompt p specifies instrumentation only and is encoded by frozen T5Gemma. Query and matched-baseline speech are encoded by frozen wav2vec2-ER; intensity α interpolates in SER space before factorized V/A tokens are gated and fed to a dedicated mood cross-attention adapter.

2. METHOD

2.1. Overview

Given a speech utterance x_s and an emotion-free instrument prompt p, SEAM conditions a frozen text-to-music DiT through separate text and affect pathways. A frozen T5Gemma encoder maps p to $\mathbf{C}_{\text{text}}(p)$, while a lightweight affect conditioner extracts $N = 32$ speech-derived mood tokens, consisting of 16 valence and 16 arousal tokens, which are injected through dedicated mood cross-attention modules.

2.2. Speech Affect Encoding and Mood Tokens

A frozen wav2vec2-base SER model [18, 19] encodes x_s. Its final hidden states are mean-pooled to obtain $\mathbf{z} \in R^{768}$ and projected as $\mathbf{h} = f_{\text{tr}}(\mathbf{z})$. An auxiliary head predicts continuous valence and arousal:

$$[\hat{v}, \hat{a}]^\top = \tanh(W_{\text{va}}\mathbf{h}). \tag{1}$$

Independent MLPs map $\mathbf{h}$ to valence and arousal latents $\mathbf{u}_v, \mathbf{u}_a \in R^{768}$, which are aligned with music-side harmony and energy representations, respectively (Sec. 2.4). Separate token heads expand them into 16 DiT-width tokens each:

$$\mathbf{T}_{\text{mood}} = [\mathbf{T}_v; \mathbf{T}_a] \in R^{32\times 768}. \tag{2}$$

2.3. Mood Cross-Attention and Intensity Control

We add a dedicated mood cross-attention module to each DiT block while preserving the original text cross-attention pathway. The valence and arousal branches are independently gated by learnable scalars g_v and g_a, while a global gain s_{mood} controls the overall strength of speech-derived conditioning.

For continuous affect control, let $\mathbf{z}_s$ and $\mathbf{z}_b$ denote the SER embeddings of the target and matched baseline utterances. We interpolate between them as

$$\mathbf{z}(\alpha) = \mathbf{z}_b + \alpha(\mathbf{z}_s - \mathbf{z}_b), \qquad \alpha \in \{0, 0.25, 0.50, 0.75, 1.00\}, \tag{3}$$

where α controls the displacement from baseline toward the target affect.

The baseline-relative mood tokens are

$$\Delta\mathbf{T}_k(\alpha) = \mathbf{T}_k(\mathbf{z}(\alpha)) - \mathbf{T}_k(\mathbf{z}_b), \qquad k \in \{v, a\}, \tag{4}$$

and the injected tokens are

$$\widetilde{\mathbf{T}}_k(\alpha) = s_{\text{mood}}\, \sigma(g_k)\, \Delta\mathbf{T}_k(\alpha). \tag{5}$$

During training, a paired neutral utterance is used as the baseline when available; otherwise, absolute mood tokens are used.

2.4. Training Objective

For speech–music alignment, we use harmony-related features (chroma and key) for valence and energy-related features (RMS, onset, and tempo) for arousal, following [11]. Music2Emo [20] provides music-side VA estimates for constructing cross-modal similarity targets.

The overall objective is

$$\mathcal{L} = \lambda_d\mathcal{L}_{\text{diff}} + \lambda_m\mathcal{L}_{\text{map}} + \lambda_r\mathcal{L}_{\text{rank}} + \lambda_{\text{va}}\mathcal{L}_{\text{VA}}. \tag{6}$$

Here, $\mathcal{L}_{\text{diff}}$ is the rectified-flow velocity loss under joint text and mood conditioning. For each axis $k \in \{v, a\}$, $\mathcal{L}_{\text{map}}$ aligns the speech latent $\mathbf{u}_k$ with the corresponding music-side representation using the target similarity

$$s_k = \exp\left(-\frac{(y_k - y_{m,k})^2}{2\sigma^2}\right), \qquad \sigma = 0.5, \tag{7}$$

where y_k is the speech-side target and $y_{m,k}$ is the corresponding Music2Emo estimate. Finally, $\mathcal{L}_{\text{rank}}$ preserves pairwise VA ordering, while $\mathcal{L}_{\text{VA}}$ supervises continuous speech-side VA prediction.

Table 1. Loss ablation, IEMOCAP S5 Task A. Same 60 clips, three seeds, 8 s / 20 steps. Each other row zeros one coefficient in Eq. (6).

Method	VA↓	VCCC	ACCC	Acc.
w/o $\mathcal{L}_{\text{rank}}$	1.09	0.17	0.11	0.28
w/o $\mathcal{L}_{\text{map}}$	1.16	**0.19**	0.13	0.33
w/o $\mathcal{L}_{\text{VA}}$	1.14	0.09	0.06	0.33
Full	**1.07**	0.13	**0.18**	**0.39**

Table 2. Task A transfer. IEMOCAP target = official (V, A); ESD = tag Acc. only. n=180/corpus.

Corpus	Method	VA↓	VCCC↑	ACCC↑	Acc.↑
IEMOCAP	SEAM	**0.81**	**0.27**	**0.29**	**0.45**
	Numeric VA text	0.92	0.01	0.01	0.27
	Lexical text	0.96	0.02	0.02	0.13
ESD	SEAM	–	–	–	**0.88**
	Lexical text	–	–	–	0.33

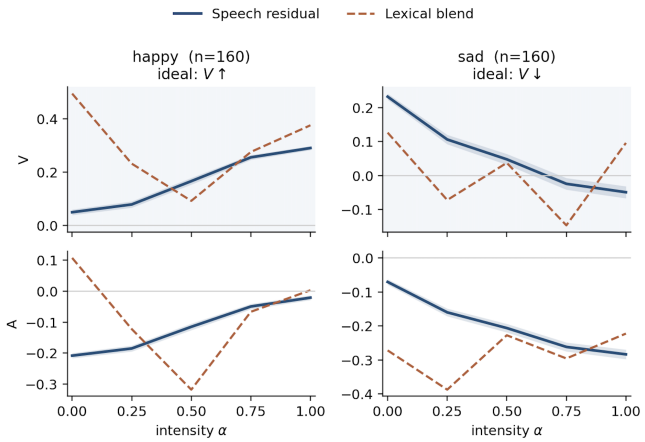


Fig. 2. Valence (V) and arousal (A) trajectories over affect intensity α for happy and sad conditions (n=160 each), comparing SEAM speech-residual steering with lexical blending.

To examine the contribution of each auxiliary objective, we conduct loss ablations by removing $\mathcal{L}_{\text{rank}}$, $\mathcal{L}_{\text{map}}$, or $\mathcal{L}_{\text{VA}}$ while keeping all other training components fixed. As reported in Table 1, the full objective yields the best overall affect-alignment performance, indicating that cross-modal feature mapping, VA-order preservation, and direct continuous VA supervision provide complementary signals.

3. EXPERIMENTS

3.1. Datasets

We train SEAM using speech from IEMOCAP [7] and ESD [21], paired with Jamendo music. For IEMOCAP, the official `EmoVal` and `EmoAct` ratings on a 1–5 scale are mapped to $[-1, 1]$. Session 5 is held out from speech-VA training and used for evaluation. For ESD, speakers `0017–0020` are held out. IEMOCAP provides utterance-level continuous VA annotations, whereas ESD is evaluated using emotion-class VA prototypes. Emotion-related terms are removed from Jamendo metadata to prevent lexical affect leakage.

3.2. Implementation Details

The SA3 medium backbone, VAE, and T5Gemma encoder are frozen, and only the affect conditioner and mood cross-attention modules are trained using AdamW with a learning rate of 10^{-4}, batch size 8, and seed 42. Automatic evaluation uses 8 s generations with 20 sampling steps and classifier-free guidance of 1.5, while human evaluation uses 30 s samples. All systems use the same emotion-free instrument prompt: `piano, violin string, and drum`.

3.3. Tasks and Metrics

Generated music is evaluated using Music2Emo [20], which predicts valence and arousal (V, A).

Task A: Emotion Alignment. We evaluate 20 utterances per emotion for three representative emotions (happy, sad, and angry) in each corpus, using three diffusion seeds. For IEMOCAP, the target is the official utterance-level (V, A); for ESD, we use the corresponding emotion-class VA prototype. We report Euclidean VA distance, valence and arousal concordance correlation coefficients (CCC), and 3-way nearest-prototype accuracy.

Task B: Intensity Control. For each utterance and diffusion seed, we generate music at $\alpha \in \{0, 0.25, 0.50, 0.75, 1.00\}$ and analyze the resulting valence–arousal trajectories.

SEAM interpolates between baseline and target speech representations using Eq. (3), while the lexical baseline uses the same α grid in text space.

3.4. Baselines

All methods use the same frozen SA3 checkpoint, instrument prompt, sampling configuration, and diffusion seeds.

Lexical Emotion Prompting. The speech-conditioning pathway is removed and affect is specified through text. For Task A, the corresponding emotion word is appended to the prompt. For Task B, affect intensity is controlled using

$$s(\alpha) = \text{“}p \text{ with } (1-\alpha) \text{ of neutral and } \alpha \text{ of } \{\text{e}\}\text{.”} \qquad (8)$$

Numeric VA Text (IEMOCAP only). This baseline directly inserts the official Session 5 $(v, a) \in [-1, 1]$ values into the prompt:

`piano, violin string, and drum, valence` v`,` `arousal` a`.`

This provides explicit continuous affect values without speech conditioning.

3.5. Human Evaluation

We conduct a Prolific listening study with n=43 participants using 30 s IEMOCAP Session 5 samples from happy, sad, and angry conditions. Listeners first hear the reference speech and then the generated music; system identities are hidden and presentation order is randomized. Each participant completes 18 Task 1 ratings (3 emotions × 3 systems × 2 clips) and 30 Task 2 ratings (3 emotions × 2

Table 3. Task 2 S1 proximity by emotion. α=0 is the $S2$ (baseline) endpoint and α=1 is the $S1$ (target) endpoint; higher C_{S1} means the clip is perceived as closer to $S1$. n=43 listeners. $\Delta C = C_{S1}(\alpha{=}1) - C_{S1}(\alpha{=}0)$ is the perceived shift from $S2$ toward $S1$.

Emotion	Model	α=0 ($S2$)	α=0.25	α=0.5	α=0.75	α=1 ($S1$)	ΔC
Happy	Ours	$0.21{\pm}0.05$	$0.25{\pm}0.05$	$0.59{\pm}0.05$	$0.32{\pm}0.04$	$0.63{\pm}0.05$	$\mathbf{0.42{\pm}0.07}$
	SA3 lexical	$0.65{\pm}0.05$	$0.38{\pm}0.05$	$0.65{\pm}0.05$	$0.27{\pm}0.05$	$0.33{\pm}0.04$	$-0.32{\pm}0.06$
Sad	Ours	$0.17{\pm}0.03$	$0.39{\pm}0.05$	$0.85{\pm}0.04$	$0.84{\pm}0.04$	$0.84{\pm}0.04$	$\mathbf{0.67{\pm}0.05}$
	SA3 lexical	$0.44{\pm}0.05$	$0.65{\pm}0.05$	$0.69{\pm}0.05$	$0.65{\pm}0.05$	$0.88{\pm}0.04$	$0.44{\pm}0.06$
Angry	Ours	$0.35{\pm}0.04$	$0.55{\pm}0.05$	$0.69{\pm}0.04$	$0.52{\pm}0.04$	$0.61{\pm}0.05$	$\mathbf{0.26{\pm}0.06}$
	SA3 lexical	$0.52{\pm}0.05$	$0.62{\pm}0.05$	$0.55{\pm}0.05$	$0.48{\pm}0.04$	$0.62{\pm}0.05$	$0.10{\pm}0.07$

Table 4. Task 1 emotion alignment. Each listener rated two clips per cell; we average within listener then across listeners. Likert 1–5 ($n = 43$ listeners).

Emotion	Metric	Ours	SA3 lexical	SA3 VA
Happy	Align	$\mathbf{2.85{\pm}0.15}$	$2.64{\pm}0.17$	$1.90{\pm}0.12$
	Quality	$\mathbf{3.63{\pm}0.14}$	$3.57{\pm}0.14$	$3.52{\pm}0.14$
	Adhere	$\mathbf{3.81{\pm}0.15}$	$3.35{\pm}0.15$	$3.72{\pm}0.18$
Sad	Align	$\mathbf{3.71{\pm}0.13}$	$3.29{\pm}0.14$	$2.86{\pm}0.14$
	Quality	$\mathbf{4.16{\pm}0.12}$	$3.93{\pm}0.13$	$3.69{\pm}0.12$
	Adhere	$\mathbf{4.41{\pm}0.13}$	$3.63{\pm}0.13$	$3.95{\pm}0.13$
Angry	Align	$\mathbf{3.15{\pm}0.16}$	$2.58{\pm}0.16$	$3.02{\pm}0.16$
	Quality	$3.83{\pm}0.14$	$3.65{\pm}0.14$	$\mathbf{3.88{\pm}0.12}$
	Adhere	$3.88{\pm}0.13$	$3.74{\pm}0.15$	$\mathbf{3.90{\pm}0.14}$

systems $\times$ 5 intensity levels). We report mean $\pm$ SEM across listeners.

Task 1: Emotion Alignment. Listeners rate each generated clip on three 5-point Likert scales: emotion alignment with the reference speech, audio quality, and prompt adherence. We compare SEAM with SA3 lexical prompting and SA3 numeric-VA text conditioning.

Task 2: Intensity Control. Each trial uses a target utterance S1 and an opposite-emotion utterance S2. Music is generated over the same α ladder for SEAM (Eq. (3)) and lexical blending. Listeners rate whether the generated music is closer to S1 or S2 on a 5-point scale, where $r = 1$ indicates much closer to S1 and $r = 5$ indicates much closer to S2. We convert this rating into S1 proximity:

$$C_{\mathrm{S1}} = \frac{5 - r}{4} \in [0, 1], \tag{9}$$

where larger values indicate greater perceptual similarity to the target S1 affect. We additionally report

$$\Delta C = C_{\mathrm{S1}}(\alpha = 1) - C_{\mathrm{S1}}(\alpha = 0), \tag{10}$$

which measures the perceived endpoint shift toward the target affect.

4. RESULTS

4.1. Emotion Alignment

Table 2 reports the automatic results for Task A. On IEMOCAP Session 5, SEAM achieves the lowest VA distance and positive CCC on both affective dimensions, outperforming lexical prompting and the numeric-VA text baseline, despite the latter receiving ground-truth VA values in its prompt. On held-out ESD speakers, SEAM also achieves higher emotion-prototype accuracy than lexical prompting. We treat the IEMOCAP results as primary because they use utterance-level continuous VA targets, whereas ESD provides only class-level VA prototypes.

Human ratings in Table 4 show the same overall trend. SEAM achieves the highest perceived emotion alignment for all three evaluated emotions, with the largest improvement for sad. Audio quality and prompt adherence are also comparable to or higher than the two SA3 baselines across the evaluated conditions.

4.2. Intensity Control

Fig. 2 summarizes the automatic results for Task B. As α increases, SEAM raises valence for happy speech and lowers it for sad speech, following the intended affective directions from the matched baseline toward the target speech. Compared with lexical blending, speech-residual steering follows the intended valence direction more consistently for the happy and sad conditions shown in Fig. 2. Because the steering direction is derived from the baseline-to-target speech representation, SEAM produces utterance-dependent trajectories rather than applying the same text-space interpolation to every utterance. Human evaluation in Table 3 further supports this trend. SEAM starts closer to S2 at $\alpha = 0$ and generally moves toward S1 as α increases, indicating a larger perceptual shift toward the target affect than lexical blending.

Limitations. The affect trajectories are not perfectly monotonic, particularly for happy and angry, suggesting that linear interpolation in SER space does not necessarily yield perceptually linear changes in music affect. The human study covers a limited number of listeners and stimuli, and Music2Emo may not perfectly align with the IEMOCAP affect space.

5. CONCLUSION

We presented SEAM, a lightweight framework for transferring speech affect to instrumental music without explicit emotion words in the text prompt. SEAM injects factorized valence and arousal tokens into a frozen Stable Audio 3 model and uses baseline-relative SER-space steering for continuous affect control. Objective and human evaluations show that SEAM improves speech–music affect alignment over same-backbone text-based conditioning and produces larger perceptual changes toward target affect under intensity control. Future work will improve the robustness and perceptual monotonicity of continuous affect control and extend evaluation to broader speech and music domains.

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